EXPERIMENT :- 4

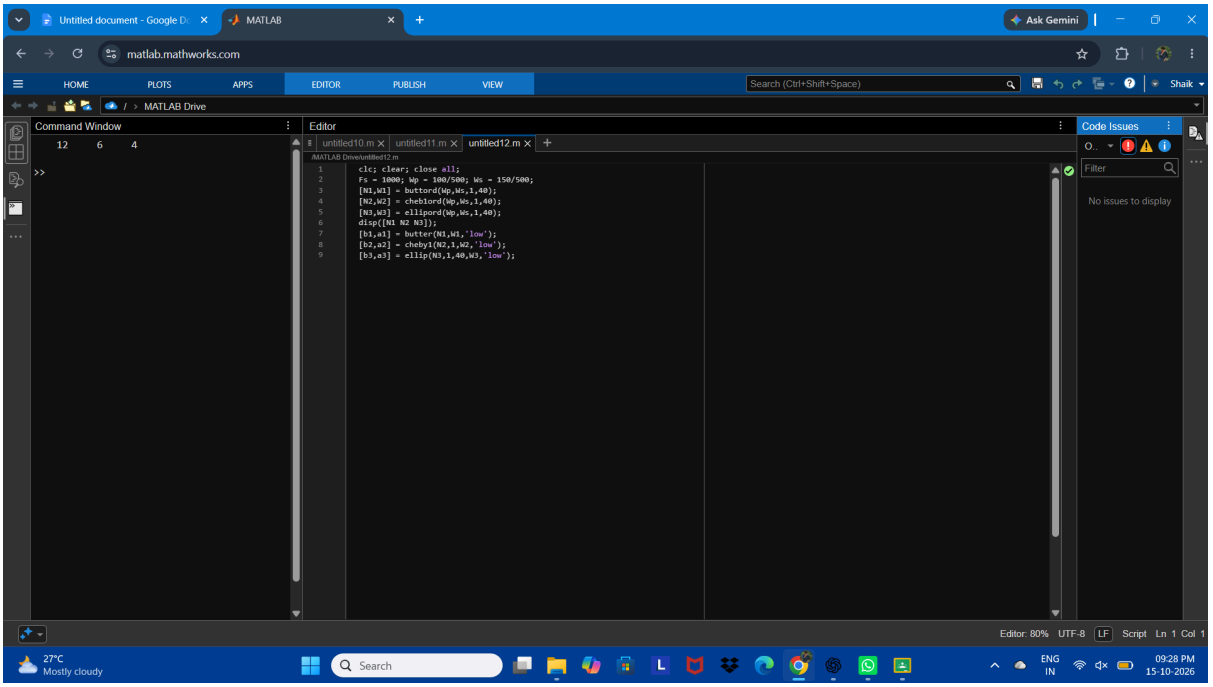
PROGRAM CODES ➖

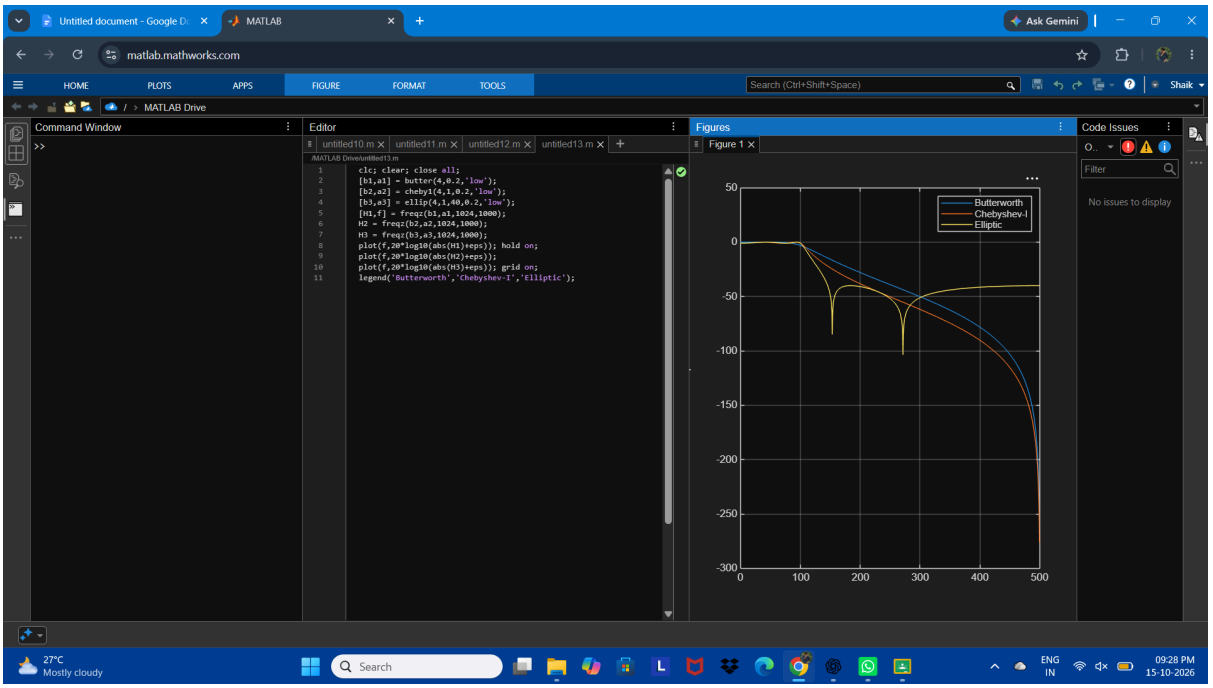
```
clc; clear; close all;
Wn = 100/500;
[b1,a1] = butter(4,Wn,'low');
[b2,a2] = cheby1(4,1,Wn,'low');
[b3,a3] = ellip(4,1,40,Wn,'low');
freqz(b1,a1,512,1000);
figure; freqz(b2,a2,512,1000);
figure; freqz(b3,a3,512,1000);
```

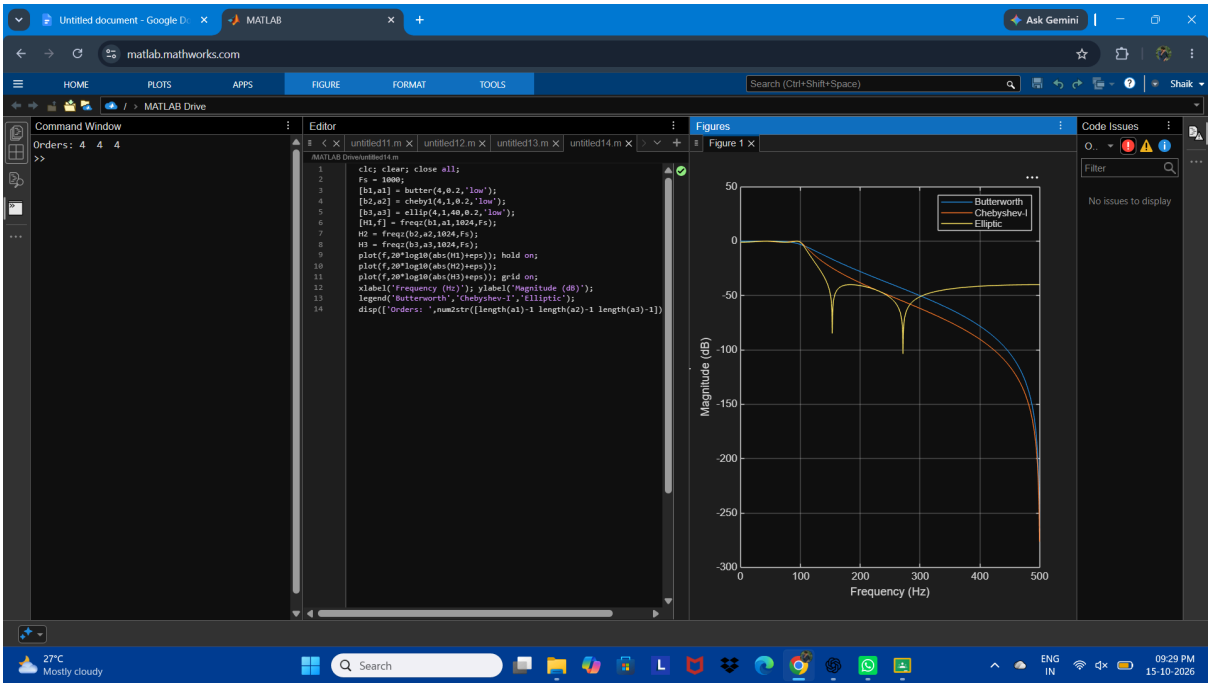
```
clc; clear; close all;
Fs = 1000; Wp = 100/500; Ws = 150/500;
[N1,W1] = buttord(Wp,Ws,1,40);
[N2,W2] = cheb1ord(Wp,Ws,1,40);
[N3,W3] = ellipord(Wp,Ws,1,40);
disp([N1 N2 N3]);
[b1,a1] = butter(N1,W1,'low');
[b2,a2] = cheby1(N2,1,W2,'low');
[b3,a3] = ellip(N3,1,40,W3,'low');
```

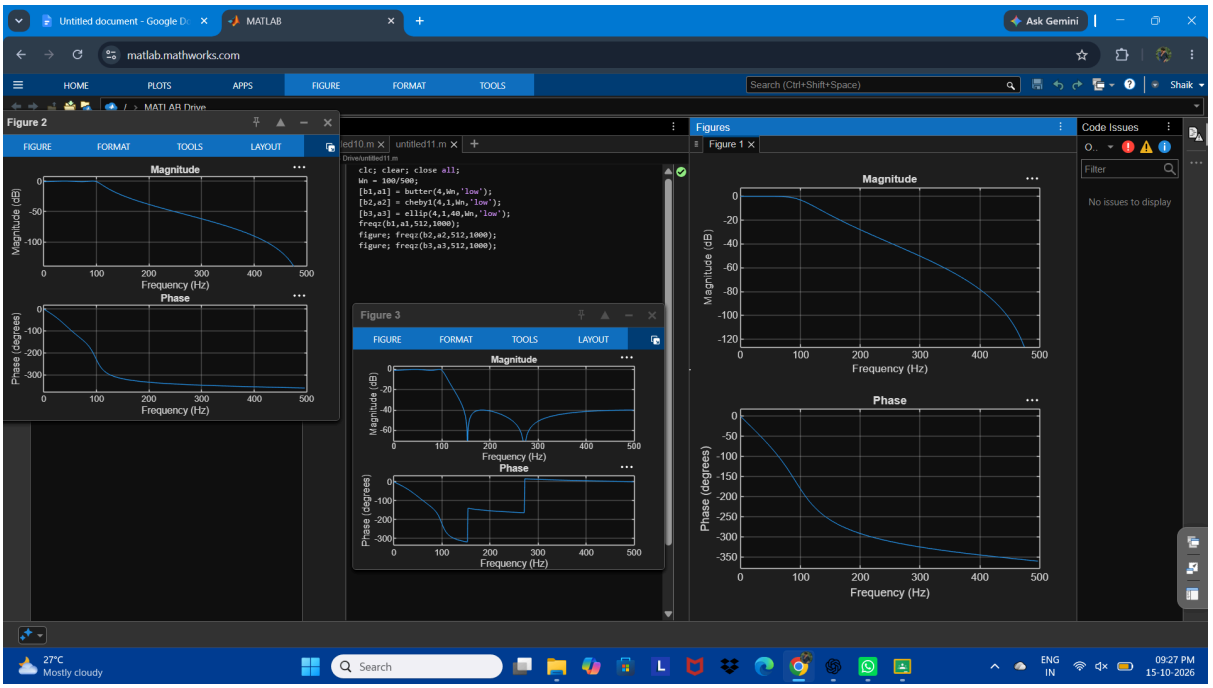
```
clc; clear; close all;
[b1,a1] = butter(4,0.2,'low');
[b2,a2] = cheby1(4,1,0.2,'low');
[b3,a3] = ellip(4,1,40,0.2,'low');
[H1,f] = freqz(b1,a1,1024,1000);
H2 = freqz(b2,a2,1024,1000);
H3 = freqz(b3,a3,1024,1000);
plot(f,20*log10(abs(H1)+eps)); hold on;
plot(f,20*log10(abs(H2)+eps));
plot(f,20*log10(abs(H3)+eps)); grid on;
legend('Butterworth','Chebyshev-I','Elliptic');
```

```
clc; clear; close all;
Fs = 1000;
[b1,a1] = butter(4,0.2,'low');
[b2,a2] = cheby1(4,1,0.2,'low');
[b3,a3] = ellip(4,1,40,0.2,'low');
[H1,f] = freqz(b1,a1,1024,Fs);
H2 = freqz(b2,a2,1024,Fs);
H3 = freqz(b3,a3,1024,Fs);
plot(f,20*log10(abs(H1)+eps)); hold on;
plot(f,20*log10(abs(H2)+eps));
plot(f,20*log10(abs(H3)+eps)); grid on;
xlabel('Frequency (Hz)'); ylabel('Magnitude (dB)');
legend('Butterworth','Chebyshev-I','Elliptic');
disp(['Orders: ',num2str([length(a1)-1 length(a2)-1 length(a3)-1])]);
```


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clc; clear; close all;
Fs = 1000; Wp = 100/500; Ws = 150/500;
[N1,W1] = buttord(Wp,Ws,1,40);
[N2,W2] = cheb1ord(Wp,Ws,1,40);
[N3,W3] = ellipord(Wp,Ws,1,40);
disp([N1 N2 N3]);
[b1,a1] = butter(N1,W1,'low');
[b2,a2] = cheby1(N2,1,W2,'low');
[b3,a3] = ellip(N3,1,40,W3,'low');
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clc; clear; close all;
[b1,a1] = butter(4,0.2,'low');
[b2,a2] = cheby1(4,1,0.2,'low');
[b3,a3] = ellip(4,1,40,0.2,'low');
[H1,f] = freqz(b1,a1,1024,1000);
H2 = freqz(b2,a2,1024,1000);
H3 = freqz(b3,a3,1024,1000);
plot(f,20*log10(abs(H1)+eps)); hold on;
plot(f,20*log10(abs(H2)+eps));
plot(f,20*log10(abs(H3)+eps)); grid on;
legend('Butterworth','Chebyshev-I','Elliptic');
Figures
Figure 1
Butterworth
Chebyshev-I
Elliptic
50
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clc; clear; close all;
Fs = 1000;
[b1,a1] = butter(4,0.2,'low');
[b2,a2] = cheby1(4,1,0.2,'low');
[b3,a3] = ellip(4,1,40,0.2,'low');
[H1,f] = freqz(b1,a1,1024,Fs);
H2 = freqz(b2,a2,1024,Fs);
H3 = freqz(b3,a3,1024,Fs);
plot(f,20*log10(abs(H1)+eps)); hold on;
plot(f,20*log10(abs(H2)+eps));
plot(f,20*log10(abs(H3)+eps)); grid on;
xlabel('Frequency (Hz)'); ylabel('Magnitude (dB)');
legend('Butterworth','Chebyshev-I','Elliptic');
disp(['Orders: ',num2str([length(a1)-1 length(a2)-1 length(a3)-1])
Figures
Figure 1
Butterworth
Chebyshev-I
Elliptic
Magnitude (dB)
Frequency (Hz)
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TOOLS
LAYOUT
Magnitude
Phase
Magnitude (dB)
Phase (degrees)
Frequency (Hz)
Wn = 100/500;
[b1,a1] = butter(4,Wn,'low');
[b2,a2] = cheby1(4,1,Wn,'low');
[b3,a3] = ellip(4,1,40,Wn,'low');
freqz(b1,a1,512,1000);
figure; freqz(b2,a2,512,1000);
figure; freqz(b3,a3,512,1000);
Figure 3
Figures
Figure 1
Code Issues
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